Lecture 12 - Shrinkage Methods

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Motivation for Shrinkage

Suppose we want to predict “group” mean.

- bias and variance tradeoff
- optimal combination of grand mean and group mean
- example: govt workers in CPS
- Ridge regression & tuning parameter
- Lasso regression

Reference: ISLR – Chapter 6.2

Motivation for Shrinkage: Example

Classic "signal processing" problem

- Given normally distributed r.v. x, and signal $s = x + \epsilon$ with normally distributed noise,

$$\begin{aligned} x &\sim N(\mu, \sigma_x^2) \\ \epsilon &\sim N(0, \sigma_\epsilon^2) \\ s = x + \epsilon &\sim N(0, \sigma_x^2 + \sigma_\epsilon^2) \end{aligned}$$

- What is $E[x|s]$? Since x, s are joint normal ("normal update"):

$$E[x|s] = \underbrace{E[x]}_{\text{prior}} + r_{x,s} \underbrace{(s - E[s])}_{\text{signal}}$$

where $r_{x,s} = cov[x,s]/var[s] = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\epsilon^2}$ = regression coefficient (x on $(1, s)$)

$$\Rightarrow E[x|s] = \mu + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\epsilon^2}(s - \mu) = (1 - \lambda)\mu + \lambda s$$

where $\lambda = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\epsilon^2}$ is "signal to total variance" ratio. We "shrink" the signal s by a factor λ.

Framework: *Group-specific Means*

- Setup: y_{vi}= observed outcome for observation i in group $v = 1, 2, \ldots V$. DGP has a "group" structure:

$$y_{vi} = \alpha + \beta_v + \epsilon_{vi} \tag{1}$$

- OLS: we could estimate a model like

$$y_{vi} = \sum_{v=1}^{V} D_{vi}\beta_v + u_{vi} \tag{2}$$

  where D_{vi}= dummy for group v.
  - Sample has N elements, with N_v observations for group v
  - Balanced sample: N_v same for each group; Unbalanced: N_v varies
- BUT: OLS (2) may be **"over-fitting"** - the $\widehat{\beta}'_v s$ will be noisy.
- The shrinkage idea: use a weighted average of grand mean $\overline{y}$ and the group mean $\overline{y}_v$:

$$\widehat{\beta}_v^{Shrink} = \theta\overline{y} + (1-\theta)\overline{y}_v \tag{3}$$

  If $\theta > 0$, we are shrinking the the prediction towards $\bar{y}$. OLS sets $\theta = 0$ (no weight on $\overline{y}$).

Assumption - Random effect model

- Under the group structure (1),

$$y_{vi} = \alpha + \beta_v + \epsilon_{vi}$$

  assume:

  1. $N_v/N = s_v$ are fixed (across draws).
  2. $\{\beta_v\}$ are random effects of groups. That is, think of β_v's as a random variable, which satisfies:

$$E[\beta_v] = \sum_v s_v\beta_v = 0,\ E[\beta_v^2] = \sum_v s_v\beta_v^2 = \sigma_v^2$$

  3. ϵ_{vi} are i.i.d. with $E[\epsilon_{vi}] = 0$ and $var[\epsilon_{vi}] = \sigma_\epsilon^2$

- We ask: what is our best estimate for the mean of a new observation from group v? Clearly, we want to get our best estimate of

$$\alpha + \beta_v$$

Shrinkage estimator for $\alpha + \beta_v$

- OLS (2) predicts the mean in group v is just $\hat{\alpha} + \hat{\beta}_v = \overline{y}_v$. Is this the best we can do?
- Consider shrinkage estimators:

$$\hat{y}_v^{\theta} := \theta \overline{y} + (1 - \theta)\overline{y}_v$$

  i.e., combine the *grand mean* and the *group-specific mean*.
- Why might we want to "shrink"?
  - if N_v is relatively small, $\overline{y}_v$ will be noisy.
  - $\overline{y}$ is a good estimate of α.
  - if $var(\beta_v) = \sigma_v^2$ (variance across groups) is small, there is information in $\overline{y}$ that we can use!

Shrinkage estimator - MSFE

- Consider estimator $\hat{y}_v^\theta := \theta\overline{y} + (1-\theta)\overline{y}_v$ of a "new" obs from group v.

$$E[(y_i - \hat{y}_v^\theta)^2 | i \in v] = \underbrace{E[(y_i - E[y_i|v])^2 | i \in v]}_{\text{irreducible}} + E[\underbrace{(E[y_i|v] - E[\hat{y}_v^\theta])^2}_{\text{expected bias}^2}] + \underbrace{var(\hat{y}_v^\theta)}_{\text{variance}} \tag{4}$$

- LHS is the mean squared forecast error for group v: $MSFE(\hat{y}_v^\theta) = E[(y_i - \hat{y}_v^\theta)^2 | i \in v]$.

Shrinkage estimator - Bias

- What is bias? $E[\overline{y}] = \alpha$; $E[\overline{y}_v] = \alpha + \beta_v \Rightarrow$

$$\begin{aligned} E[\hat{y}_v^\theta] &= E[\theta\overline{y} + (1-\theta)\overline{y}_v] \\ &= \alpha + (1-\theta)\beta_v \\ Bias(v) &= E[y|v] - E[\theta\overline{y} + (1-\theta)\overline{y}_v] \\ &= (\alpha + \beta_v) - \alpha + (1-\theta)\beta_v \\ &= \theta\,\beta_v \end{aligned} \tag{5}$$

  unbiased if $\theta = 0$. All weight on $\overline{y}_v$.
- Note β_v is the random effect for group v, with $E[\beta_v] = 0, E[\beta_v^2] = \sigma_v^2$. The expected bias-squared is

$$E_v[Bias(v)^2] = \theta^2 E[\beta_v^2] = \theta^2\sigma_v^2$$

.

Shrinkage estimator - Variance

- What about variance?

$$
\begin{aligned}
var(\hat{y}_v^\theta) =& E[(\theta\overline{y} + (1-\theta)\overline{y}_v - E[\hat{y}_v^\theta])^2] \\
=& E[(\theta(\overline{y} - \alpha) + (1-\theta)(\overline{y}_v - \alpha - \beta_v))^2] \\
=& \theta^2 var(\bar{y}) + (1-\theta)^2 var(\bar{y}_v) \\
& + 2\theta(1-\theta) cov(\bar{y}, \bar{y}_v) \qquad (6)
\end{aligned}
$$

- One step at a time:
    1. $var(\bar{y})$:

$$
\overline{y} = \frac{1}{N}\sum_v \sum_{i \in v} y_{vi} = \frac{1}{N}\sum_v \sum_{i \in v} (\alpha + \beta_v + \varepsilon_{vi}) = \alpha + \frac{1}{N}\sum_v \sum_{i \in v} \varepsilon_{vi}
$$

Recall we have assumed $E[\beta_v] = \sum_v s_v \beta_v = 0$ and the shares $\{s_v\}$ are held fixed. Thus

$$
var[\overline{y}] = \sigma_\epsilon^2 / N.
$$

Shrinkage estimator - Variance

2. $var(\bar{y}_v)$:

$$\bar{y}_v = \frac{1}{N_v}\sum_{i\in v} y_{vi} = \alpha + \beta_v + \frac{1}{N_v}\sum_{i\in v}\varepsilon_{vi}$$
$$\Rightarrow var[\bar{y}_v] = \sigma_\epsilon^2/N_v$$

3. $cov(\bar{y}, \bar{y}_v)$: Note $\bar{y} - E[\bar{y}] = \frac{N_v}{N}(\frac{1}{N_v}\sum_{i\in v}\varepsilon_{vi}) + \frac{1}{N}\sum_{v'\neq v}\sum_{i\in v'}\varepsilon_{vi}$ and $\bar{y}_v - E[\bar{y}_v] = \frac{1}{N_v}\sum_{i\in v}\varepsilon_{vi}$

$$cov(\bar{y}, \bar{y}_v) = \frac{N_v}{N}\frac{\sigma_\epsilon^2}{N_v} = \frac{\sigma_\epsilon^2}{N} = var[\bar{y}]$$

- Plugging the above into (6),

$$var(\hat{y}_v^\theta) = \theta^2 var(\bar{y}) + (1-\theta)^2 var(\bar{y}_v) + 2\theta(1-\theta)var(\bar{y})$$
$$= (2\theta - \theta^2)\frac{\sigma_\epsilon^2}{N} + (1-\theta)^2\frac{\sigma_\epsilon^2}{N_v}.$$

Shrinkage: Minimize bias+variance

- We want to minimize "$v + b$" $= E[bias^2] + variance$ for each group v:

$$\begin{aligned} min_\theta &= E[Bias(v)^2] + var(\hat{y}_v^\theta) \\ &= \theta^2\sigma_v^2 + (2\theta - \theta^2)\frac{\sigma_\epsilon^2}{N} + (1-\theta)^2\frac{\sigma_\epsilon^2}{N_v} \\ &= \frac{\sigma_\epsilon^2}{N}\left(\theta^2\kappa + 2\theta - \theta^2 + (1-\theta)^2\frac{1}{s_v}\right) \end{aligned} \tag{7}$$

  in which $\kappa = \frac{\sigma_v^2}{\sigma_\epsilon^2}N$, and $s_v = \frac{N_v}{N}$.

- FOC w.r.t. θ:

$$\frac{\partial "v+b"}{\partial\theta} \propto \theta\kappa + 1 - \theta - (1-\theta)\frac{1}{s_v} = 0$$

$$\Rightarrow \theta_v^* = \frac{1/s_v - 1}{\kappa - 1 + 1/s_v} = \frac{N - N_v}{N + (\kappa - 1)N_v}$$

- SOC:

$$\frac{\partial^2("v+b")}{\partial\theta^2} = \frac{2\sigma_\epsilon^2}{N}\left(\kappa - 1 + \frac{1}{s_v}\right) > 0$$

Shrinkage: Minimize bias+variance

$$\theta_v^* = \frac{N - N_v}{N + (\kappa - 1)N_v}, \; \kappa = \frac{\sigma_v^2}{\sigma_\epsilon^2 / N} \tag{8}$$

$$= \frac{1}{1 + \kappa/(\frac{1}{s_v} - 1)}$$

$$\hat{y_v}^{\theta *} = \theta_v^* \bar{y} + (1 - \theta_v^*)\bar{y}_v$$

- Observations: $\theta_v^* \downarrow$ as $\kappa \uparrow$.
  1. if $\kappa = 0$ ($i.e.$ $\sigma_v^2 = 0$) then $\theta_v^* = 1$ – use grand mean!
  2. as $\kappa \to \infty$, $\theta_v^* \to 0$
  3. if $\kappa = 1$ ($\sigma_v^2 = \frac{\sigma_\epsilon^2}{N}$), $\theta_v^* = 1 - s_v \downarrow$ as $s_v \uparrow$.
- We can easily estimate $\frac{\sigma_\epsilon^2}{N}$ so the only unknown is σ_v^2.

Old way: estimate σ_v via OLS

How can we proceed? The "old way" is to estimate σ_v^2 from the OLS model (2):

$$y_{vi} = \alpha + D_i'\beta + \epsilon_{vi}$$

where D_i is a vector of dummies for group membership (V rows).

- instead of dropping one dummy, estimate with the restriction that $\sum_v (N_v/N)\beta_v = 0$ (restricted OLS). This gets the right constant and imposes our assumption that $E[\beta_v] = 0$.
- Get $\hat{\beta}_v$ and estimate $\sum_v s_v \times \hat{\beta}_v^2$. This is too big because $\hat{\beta}_v$ is noisy!
- But we know the sampling error $\hat{\sigma}^2_{\beta_v} = \sum_v s_v(\hat{\beta}_v - \hat{\bar{\beta}}_v)^2$, so we subtract off $\sum_v s_v \hat{\sigma}^2_{\beta_v}$ (adjusting for sampling variance).

$$\sum_v s_v \hat{\beta}_v^2 = \hat{\sigma}^2_{\beta_v} + (1+\theta)^2 \underbrace{\sigma_v^2}.$$

"Tuning" process: searching over σ_v^2

- The unknown $\sigma_v^2 = var(\beta_v)$ may be viewed as a tuning parameter.
- The tuning process involves finding the optimal value of the regularization parameter to minimize MSE (or other measures of errors). This is typically done with cross validation.
  1. select a training sample and holdout sample.
  2. estimate $\bar{y}$ and $\bar{y}_v$ in the training sample
  3. choose a value of $\sigma_v \Rightarrow \theta_v^*$ according to (6). Note: different θ_v^* for each group
     - for observation i in the holdout sample in group $v(i)$, prediction is: $\theta^*_{v(i)}\bar{y} + (1 - \theta^*_{v(i)})\bar{y}_{v(i)}$
  4. Form $MSE(\sigma_v)$ in the holdout/test sample:

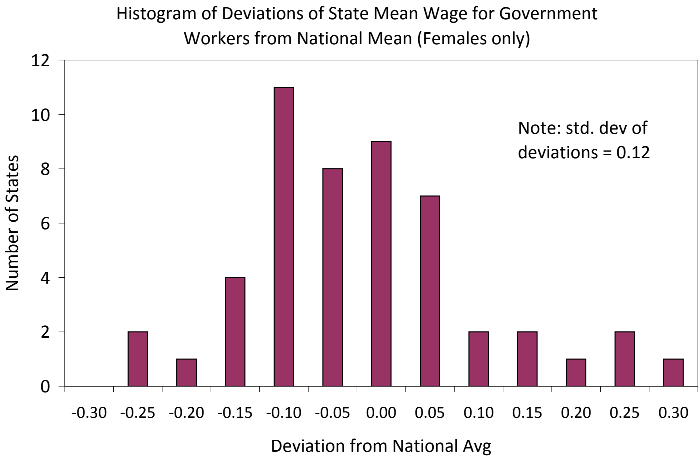
$$MSE(\sigma_v) = \frac{1}{N_H} \sum_{i \in H} (y_i - \theta^*_{v(i)}\bar{y} - (1 - \theta^*_{v(i)})\bar{y}_{v(i)})^2$$

  5. Choose σ_v^* that minimizes MSE(σ_v).

CPS

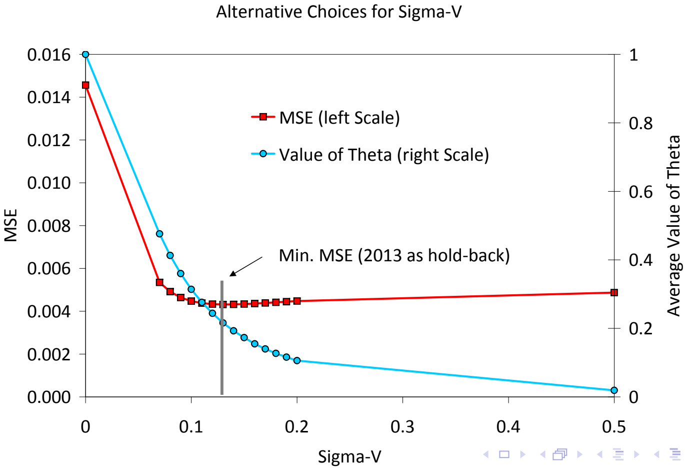
- We have CPS sample for 2012 and 2013, with data on female workers in the government sector. We are interested in forming predictions for wages for government workers in different states. The sample sizes per state range from 140 for Idaho to 1350 for California.
- A few relevant stats:
  - N=14,666 – excluding 376 obs from DC (state=53) which is outlier
  - grand mean of log hourly wage is 2.958 ($19.26/hr).
  - deviations of state from national average: std dev = 0.12 (crude estimate of σ_v^2)

CPS

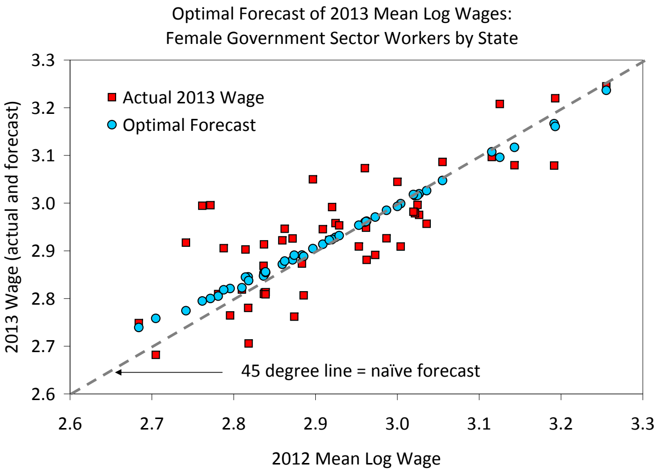

Histogram of Deviations of State Mean Wage for Government Workers from National Mean (Females only)
Note: std. dev of deviations = 0.12
Number of States
0
2
4
6
8
10
12
-0.30
-0.25
-0.20
-0.15
-0.10
-0.05
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Deviation from National Avg

CPS - Select σ_v^2

Tuning parameter σ_v^2: select the optimal σ_v^2 s.t. MSE is minimized in the test sample:



CPS - Shrinkage Estimates



Ridge Regression

- A more general version of the same class of problems: regression model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + ... + \beta_K x_{Ki} + \epsilon_i$$

  We are worried that the estimated $\beta' s$ will be "over-fit".
- Idea: penalize large $\hat{\beta}_j' s$
- Ridge regression objective:

$$min_\beta \sum_i (y_i - \beta_0 - \beta_1 x_{1i} - ... - \beta_K x_{Ki})^2 + \lambda \sum_{j=1}^{K} \beta_j^2$$

  Notes:

  1. don't penalize the constant
  2. setting $\lambda = 0$ is the usual OLS regression model
  3. larger values of λ penalize larger $\hat{\beta}_j' s$ more and shrink $\hat{\beta}_j' s$.
  4. have to be careful to scale the $x_j' s$
  5. in the group dummy case, all $x' s$ are already on the same scale; otherwise rescale e.g.: $x_{1i} = (x_{1i} - \overline{x}_1)/\sigma(x_{1i})$ ("z-scores")

Ridge (Vector Notation)

$$min_{\beta} \sum_{i} (y_i - \beta_0 - \beta_1 x_{1i} - ... - \beta_K x_{Ki})^2 + \lambda \sum_{j=1}^{K} \beta_j^2$$

- write in vector notation as

$$min\ (y - \beta_0 - X\beta)'(y - \beta_0 - X\beta) + \lambda\beta'\beta$$

- FOC for the coefficients β

$$-2X'(y - \beta_0 - X\beta) + 2\lambda\beta = 0$$

$$(X'X + \lambda I)\beta = X'(y - \beta_0)$$

$$\Rightarrow \hat{\beta} = (X'X + \lambda I)^{-1} X'(y - \beta_0)$$

Ridge - constant $\widehat{\beta}_0$

- What about $\widehat{\beta}_0$ (not penalized)?

$$min\ (y - \beta_0 - X\beta)'(y - \beta_0 - X\beta) + \lambda\beta'\beta$$

- So FOC is:

$$-1'(y - \widehat{\beta}_0 - X\widehat{\beta}) = 0$$

where 1 is the column vector of 1's, i.e.

$$\sum_i (y_i - \widehat{\beta}_0 - x_{1i}\widehat{\beta}_1 - x_{2i}\widehat{\beta}_2 - ...x_{Ji}\widehat{\beta}_J) = 0$$

- But if the $x'_{ji}s$ are rescaled to have mean 0 then $\widehat{\beta}_0 = \overline{y}$.

Ridge - $\hat{\beta}(\lambda)$

- Back to the other $\beta's$, assuming $\widehat{\beta}_0 = \bar{y}$

$$\widehat{\beta}(\lambda) = (X'X + \lambda I)^{-1} X'(y - \bar{y})$$

- What if the $x'_{ji}s$ are just group dummies? $x_{vi} = x_{vi}^2 = 1[i \in v]$. Then $(X'X + \lambda I)$ is diagonal with (v, v) element:

$$\sum_i x_{vi}^2 + \lambda = N_v + \lambda$$

and so

$$\widehat{\beta}_v(\lambda) = \frac{\sum_i x_{vi}(y_i - \bar{y})}{N_v + \lambda} = \frac{N_v \bar{y}_v - N_v \bar{y}}{N_v + \lambda} = \frac{\bar{y}_v - \bar{y}}{1 + \lambda / N_v}$$

- We know $\widehat{\beta}_0^{OLS} = \bar{y}$, and $\widehat{\beta}_0^{OLS} + \widehat{\beta}_v^{OLS} = \bar{y}_v$, so

$$\widehat{\beta}_v(\lambda) = \frac{\widehat{\beta}_v^{OLS}}{1 + \lambda / N_v} < \hat{\beta}_v^{OLS} \text{ with } \lambda > 0 \tag{9}$$

Ridge - $\hat{\beta}(\lambda)$

- So if $x_{ji}'s$ are group dummies ridge "shrinks" each OLS coefficient toward 0, (by a factor that depends on N_v).

- The forecast of a new observation from group v based on Ridge estimator is:

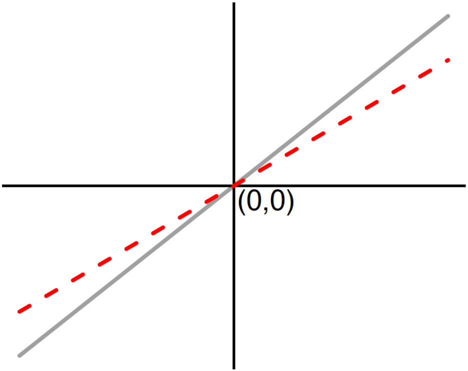
$$\hat{\beta}_0 + \widehat{\beta}_v(\lambda) = \overline{y} + \frac{\overline{y}_v - \overline{y}}{1 + \lambda / N_v} = \frac{\overline{y}_v + \frac{\lambda}{N_v}\overline{y}}{1 + \frac{\lambda}{N_v}}$$

which is a weighted average of $\overline{y}_v$ and $\overline{y}$, with weight on $\overline{y}$ of: $\frac{\frac{\lambda}{N_v}}{1+\frac{\lambda}{N_v}} = \frac{1}{1+N_v/\lambda}$.

Ridge - Visualization

$$\widehat{\beta}(\lambda) = (X'X + \lambda I)^{-1} X'(y - \overline{y})$$

Ridge



Lasso (L1 Penalty)

- **Lasso** is a penalized regression in which the penalty is based on the absolute value of the coefficients. Recall Ridge objective is:

$$min_\beta \sum_i (y_i - \beta_0 - \beta_1 x_{1i} - ... - \beta_K x_{Ki})^2 + \underbrace{\lambda \sum_{j=1}^{K} \beta_j^2}$$

  This will "shrink" coefficients toward 0, but won't eliminate them completely.

- Lasso penalizes $|\beta_j|$ instead of β_j^2 :

$$min_\beta \sum_i (y_i - \beta_0 - \beta_1 x_{1i} - ... - \beta_K x_{Ki})^2 + \underbrace{\lambda \sum_{j=1}^{K} |\beta_j|}$$

Lasso (L1 Penalty)

- In general the solution to the Lasso minimization is to set some of the coefficients to 0 – thus Lasso is a "model selection" algorithm. The reason why some coefficients are set to 0 can be seen from the nature of the Lasso objective function:

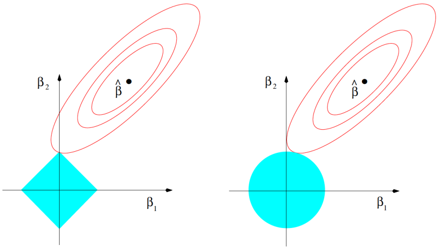
$$\begin{aligned} L(\beta) &= \sum_i (y_i - \beta_0 - \beta_1 x_{1i} - ... - \beta_K x_{Ki})^2 + \lambda \sum_{j=1}^{K} |\beta_j| \\ &= RSS(\beta) + P(\beta) \end{aligned}$$

  Consider the nature of this function, focusing on variation w.r.t. β_1.

  - The function $RSS(\beta)$ is quadratic in β_1, whereas the penalty function $P(\beta)$ is a "check" function.
  - If $\partial RSS(\beta)/\partial \beta_1$ evaluated at $\beta_1 = 0$ is not too big, it will be optimal to set $\beta_1 = 0$. By comparison, the ridge penalty function is also quadratic in β, so there is a smooth tradeoff between RSS and P.

Equivalent Representations

- Ridge: $\hat{\beta}^{Ridge} = argmin_{\beta} \|y_i - x_i' \beta\|^2 \; s.t. \underbrace{\sum_j |\beta_j|^2 \leq t}_{\sim L2-\text{penalty}}$
- Lasso: $\hat{\beta}^{Ridge} = argmin_{\beta} \|y_i - x_i' \beta\|^2 \; s.t. \underbrace{\sum_j |\beta_j| \leq t}_{\sim L1-\text{penalty}}$



Comparing Ridge and Lasso - Special case

- In the special case of a group design (with equal numbers of observations per group), the best estimate for the coefficient in Ridge regression for group v is:

$$\widehat{\beta}_v(\lambda) = \frac{\widehat{\beta}_v^{OLS}}{1 + \lambda/N_v} = \frac{\overline{y}_v - \overline{y}}{1 + \lambda/N_v}$$

- Lasso performs a different kind of shrinkage: groups with $\overline{y}_v$ close to the grand mean get $\widehat{\beta}_v = 0$, and groups farther away get $\widehat{\beta}_v = (\overline{y}_v - \overline{y}) - \delta$, where δ depends on the penalty λ.

Comparing Ridge and Lasso - General case

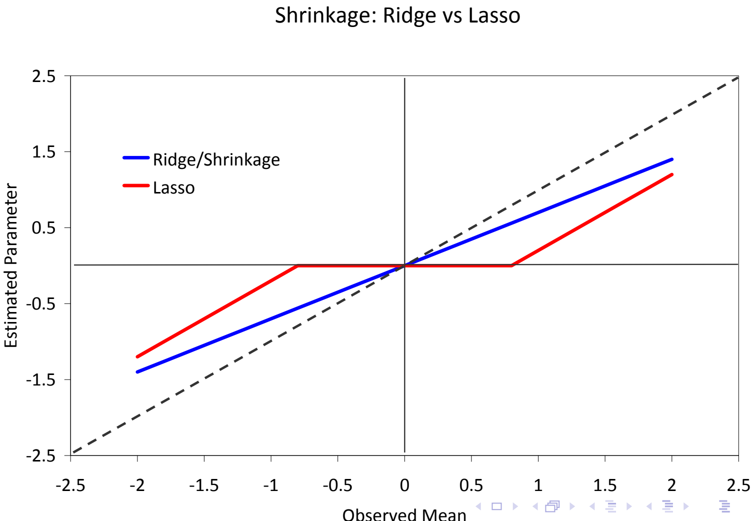
- People tend to prefer Lasso for prediction problems where the $x's$ are not necessarily group dummies. One plus: it "kicks out" weak regressors.
- Suppose $\hat{\beta}_j^{ols} > 0$, we have:

$$\hat{\beta}_j^{\text{ridge}} = \frac{\hat{\beta}_j^{ols}}{1 + \lambda/(Nv_j)} \in (0, \hat{\beta}_j^{ols})$$

$$\hat{\beta}_j^{\text{lasso}} = \hat{\beta}_j^{ols} - \lambda/(2Nv_j) \text{ if positive, } 0 \text{ otherwise}$$

where v_j =the (j,j) element of $\frac{1}{N}\sum_i x_i x_i'$. In the special case with group dummies, $v_j = N_j/N$.

Comparing Ridge and Lasso: Visualization



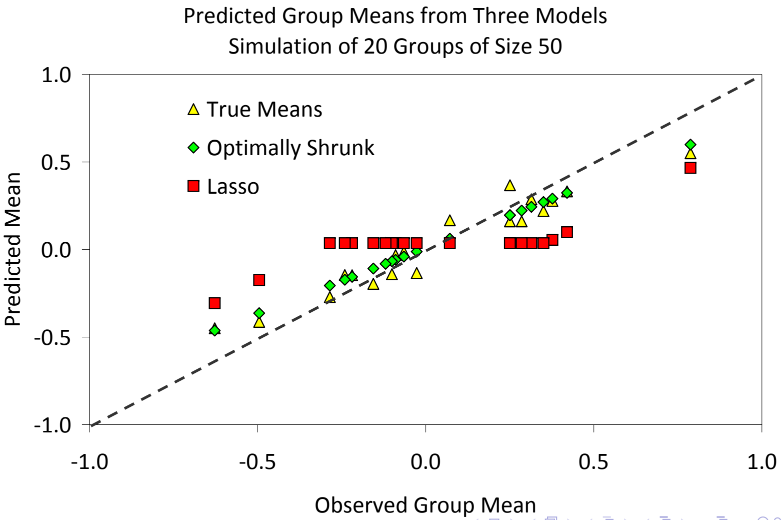
Example 1: Random Effects

Simple simulation example: a data set with 20 groups, each with 50 observations per group. DGP:

$$y_{iv} = \beta_v + \varepsilon_{iv}$$

where $\beta_v \sim N(0, \sigma_v^2)$ and $\varepsilon_{iv} \sim N(0, \sigma_\varepsilon^2)$. For this example we can calculate the "optimal" θ for shrinkage (or equivalently the optimal λ for ridge regression). We can also run the data through Lasso, using cross validation to find a "best λ" factor. Results:

Example 1: Random Effects



Implementation

How do we choose λ? Two main ways

a) cross-validation (k-fold, etc).

- choose λ, estimate penalized model on test sample

- get RMSE on holdout sample

b) penalized within-sample statistics

- choose λ, estimate penalized model on entire sample

- compare "penalized" fit statistic for entire sample